DAY 12 — Introduction to Machine Learning

Day 12 marked the beginning of **Machine Learning (ML)** concepts. This was the first day where concepts shifted from data handling to training prediction models. We explored what ML is, how algorithms work, and where they are used in real life.

🔹 1. What is Machine Learning?

Machine Learning teaches computers to learn from data.

We understood:

- **Supervised Learning**
- **Unsupervised Learning**
- **Reinforcement Learning**

The Tesla stock forecasting project falls under **Supervised Learning** with time-series characteristics.

🔸 2. Supervised Learning

We studied:

- Regression (continuous values)
- Classification (categorical values)

Examples:

- Predicting house prices (regression)
- Email spam detection (classification)

🔸 3. Popular ML Algorithms Introduced

We covered:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines
- K-Nearest Neighbors

Even though Prophet was the main model later, understanding these core algorithms built foundation.

🔸 4. Overfitting & Underfitting

We learned:

- Overfitting = too much learning
- Underfitting = too little learning
- Bias-variance tradeoff
- Regularization techniques

🔸 5. Train-Test Split

We used:

train_test_split()

to divide:

- 80% for training
- 20% for testing

This prevents models from memorizing instead of learning.

🔸 6. ML Workflow

A typical workflow includes:

1. Importing dataset
2. Cleaning
3. Feature engineering
4. Model selection
5. Training
6. Evaluation
7. Tuning
8. Deployment

Key Takeaways from Day 12

- ML algorithms are mathematical models for predictions
- Supervised learning is widely used in real applications
- Understanding data is as important as understanding the model
- ML workflow ensures systematic project development